Predicting Term-Deposit Subscription Before the Call: Leakage-Free Models and Cost-Aware Thresholds

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Introduction

Outbound telemarketing for term deposits succeeds on only about one call in nine. A bank that could estimate each client's probability of subscribing *before* dialling could call fewer people and reach more subscribers. We treat this as binary classification: given the attributes $\boldsymbol{x}$ known at dial time, estimate $P(y = 1 \mid \boldsymbol{x})$, where $y = 1$ means the client subscribed, and use the estimate to rank clients or to decide whom to call.

The public data come from a Portuguese bank's campaigns of 2008–2013 [1]. Moro, Cortez and Rita selected 22 features from about 150 and trained a neural network on earlier contacts; tested on later ones, it reached an AUC of 0.8 [1]. Later work on the same table reports much higher scores: Tékouabou et al. report 97.3% accuracy and an AUC of 0.959 with a class-membership classifier [2]. Two features of this problem make such figures hard to interpret. First, the outcome is rare, so accuracy rewards a model that always predicts "no" (88.7% here). Second, the released table includes the call duration, which is known only after the call has ended and is strongly associated with the outcome [1]. A model that uses it cannot be applied before dialling.

Our contributions address both drawbacks. We exclude duration from every main model and measure, with a separate benchmark, how much of the apparent performance it supplies. We compare models by out-of-fold ROC AUC and precision–recall AUC with bootstrap intervals, not by accuracy. And we choose the decision threshold from the costs of a wasted call and a missed subscriber, comparing the result with the trivial policies of calling everyone or nobody.

Dataset

The bank-additional-full release has 41,188 contacts, 20 inputs and the binary outcome. We removed 12 exact duplicate rows, leaving 41,176 contacts with 4,639 subscribers (11.3%), about 7.9 non-subscribers per subscriber. The inputs are client attributes (age, job, marital status, education, default, housing and personal-loan flags), the contact channel, month and weekday, the campaign history (`campaign`, `pdays`, `previous`, and `duration`), and five quarterly or monthly macroeconomic indicators (employment variation rate, consumer price and confidence indices, three-month Euribor and number employed).

Missing values are coded as the category `unknown`; we kept them as explicit indicator columns instead of imputing them. The field `pdays` uses 999 for "not previously contacted", which is not a day count: we replaced it by an indicator `never_contacted_before` and a recency score, zero when `pdays` = 999 and $1/(1 + \text{pdays})$ otherwise, and added `is_previously_engaged` from `previous`. Job and marital status were one-hot encoded and education mapped to an ordinal score with an unknown flag. Month and weekday were encoded as sine–cosine pairs, so that December lies next to January. This gives 41 pre-call features. The macroeconomic indicators are strongly correlated with each other (|r| up to 0.97; Figure 5), so individual logistic coefficients for them are not separately interpretable.

Methods

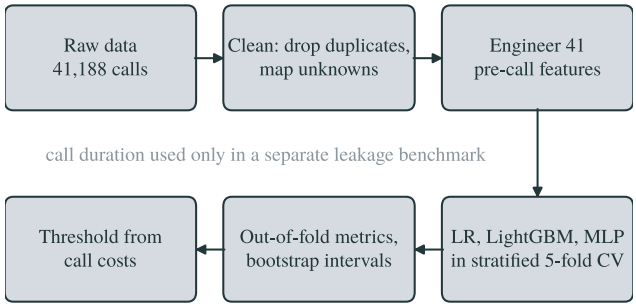


Figure 1: Method pipeline. Every reported score is out-of-fold: produced by a model that did not see that client in training.

Figure 1 summarises the pipeline. We compared three model families that handle the problem differently. Logistic regression (L2, $C = 1$, standardised inputs) is a linear, interpretable baseline. LightGBM [3], a gradient-boosted tree ensemble (1,000 trees, learning rate 0.03, 63 leaves, 80% row and column subsampling), captures non-linear effects and interactions without feature scaling. A multilayer perceptron (hidden layers of 256, 128 and 64 ReLU units, Adam, weight decay 10^{-4}, early stopping) learns its own representation [4]. Imbalance was handled by weighting the positive class by the class ratio in each model, not by resampling, so that every validation fold keeps the true class mix. Hyperparameters were set once, not tuned.

All models were evaluated by stratified five-fold cross-validation (seed 42). The out-of-fold probabilities of the five folds were concatenated into one score per client, from which we computed ROC AUC, average precision (PR-AUC), and F1 at a threshold of 0.5 and at the threshold maximising F1. Fold-to-fold standard deviations and 95% bootstrap intervals (2,000 resamples of clients) measure how far the differences between models exceed sampling noise. The duration benchmark refits LightGBM with `duration` added.

For the threshold τ, a client is called when the predicted probability is at least τ. Following the cost model fixed in our proposal, a wasted call costs \$5 and a missed subscriber \$80, so the total cost is $5\ \mathrm{FP} + 80\ \mathrm{FN}$. We compare its minimum over τ with calling nobody ($80 \times 4,639 = 371{,}120$) and calling everyone ($5 \times 36,537 = 182{,}685$).

Results and Discussion

Model	ROC AUC	PR-AUC	F1 (0.5)	F1 (t^*)
Logistic regression	0.784	0.435	0.411	0.472
MLP	0.784	0.440	0.443	0.482
LightGBM	0.784	0.450	0.472	0.486
LightGBM + duration	0.947	0.651	0.642	0.646

Table 1: Out-of-fold scores. The fold-to-fold standard deviation of ROC AUC is 0.01–0.02 and of PR-AUC 0.02; t^* is the threshold maximising F1. The last row uses call duration and is not a pre-call model.

Without duration, the three models rank clients equally well (Table 1; Figure 2a). Their ROC AUCs differ by less than 0.001, and the 95% bootstrap interval of the LightGBM–logistic difference, −0.005 to +0.006, includes zero. LightGBM is better only in the precision–recall sense: its PR-AUC exceeds that of logistic regression by 0.015 (interval 0.006 to 0.025) and that of the MLP by 0.010 (0.002 to 0.019). The gain from a non-linear model is therefore small. Most of the signal is captured by a linear function of the engineered features, which is consistent with the dominance of the macroeconomic indicators in both models (Figure 3).

Adding duration raises the ROC AUC from 0.784 to 0.947 and the PR-AUC from 0.450 to 0.651. This gap measures leakage, not a better algorithm: long calls are a consequence of interest, not a predictor available beforehand. Our pre-call AUC is close to the 0.8 that Moro et al. obtained on a later period without duration [1], and the much higher figures reported elsewhere [2] lie near our duration benchmark, which suggests that post-call information contributes to them.

The cost-optimal threshold differs from the F1-optimal one (Figure 4). Because a missed subscriber costs sixteen times a wasted call, the minimum cost lies at $\tau = 0.25$, where 41.5% of clients are called, 76% of subscribers are reached and 21% of calls succeed. The total, \$155,250, is 58% below calling nobody but only 15% below calling everyone, the better trivial policy. If every call, successful or not, costs \$5, the optimum moves to $\tau = 0.30$ and saves 16% against calling everyone. The saving depends on the assumed costs and on a threshold chosen on the same out-of-fold scores, so it is an optimistic estimate; a chronological split, training on earlier and testing on later campaigns, would test it properly.

Human agents currently decide whom to call from lists and experience; their success rate was not recorded, so we cannot compare the model with them directly. A model need not outperform an experienced agent to be useful: ranking a long list before calls begin is a task agents cannot do at scale, and the relevant benchmark is the current policy. Deployment raises further issues. Job, education and age can act as proxies for protected attributes, so contact and conversion rates should be monitored across groups. The data are personal and financial, which requires consent and an opt-out. The logistic model's coefficients allow decisions to be explained to clients. Fewer wasted calls could reduce telemarketing workload.

Conclusion

Excluding call duration, the three models rank clients equally (ROC AUC 0.784); LightGBM gives the highest PR-AUC, 0.450, by a small but significant margin. Duration alone raises the AUC to 0.947, which shows how strongly post-call information inflates reported performance. A cost-based threshold calls 41.5% of clients, reaches 76% of subscribers and saves 15% against calling everyone under the assumed costs. Validation on later campaigns, costs measured from operations and tuned hyperparameters are the next steps.

References

[1] S. Moro, P. Cortez, and P. Rita, "A data-driven approach to predict the success of bank telemarketing," *Decision Support Systems*, vol. 62, pp. 22–31, 2014, doi: 10.1016/j.dss.2014.03.001.

[2] S. C. Koumetio Tekouabou, Ş. C. Gherghina, H. Toulni, P. Neves Mata, M. N. Mata, and J. M. Martins, "A Machine Learning Framework towards Bank Telemarketing Prediction," *Journal of Risk and Financial Management*, vol. 15, no. 6, p. 269, 2022, doi: 10.3390/jrfm15060269.

[3] G. Ke *et al.*, "LightGBM: A Highly Efficient Gradient Boosting Decision Tree," in *Advances in Neural Information Processing Systems*, 2017.

[4] F. Pedregosa *et al.*, "Scikit-learn: Machine Learning in Python," *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.

Appendix

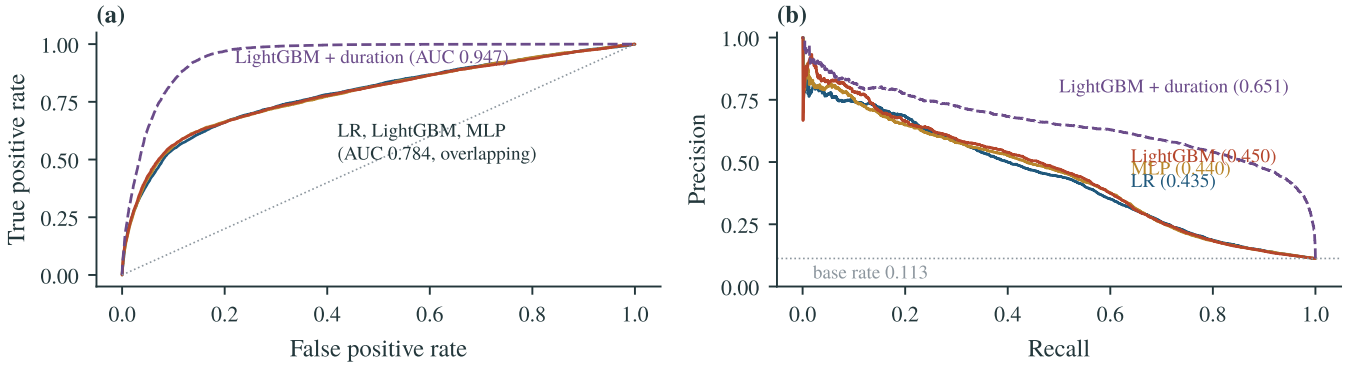


Figure 2: (a) ROC and (b) precision–recall curves of the out-of-fold scores. Without duration the three models nearly coincide; LightGBM leads slightly in precision at moderate recall.

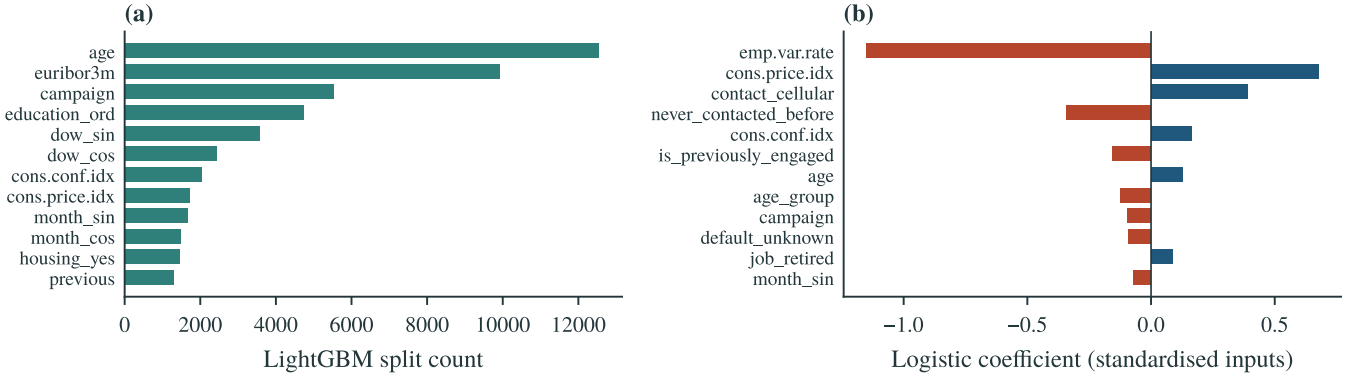


Figure 3: (a) LightGBM feature importance as the number of splits on each feature in a model refit on all data; continuous features such as age are split often, so this ranks use, not effect. (b) Largest logistic coefficients on standardised inputs. The correlated macroeconomic indicators share their effect, so their individual signs are not separately meaningful.

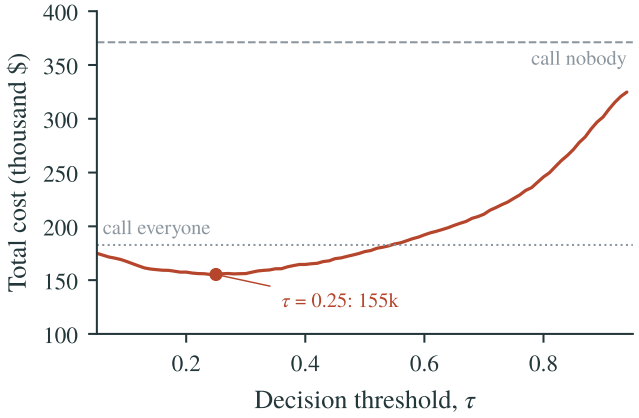


Figure 4: Total cost, 5 FP + 80 FN, against the threshold for LightGBM. Grey lines: calling nobody (dashed) and calling everyone (dotted).

AI Usage Declaration

Generative AI tools helped debug code, structure plots and improve wording; the team produced and checked all modelling decisions, runs and results. A later revision used Claude (Anthropic) to check the numbers against the saved outputs, compute the bootstrap intervals and the call-everyone cost comparison, and redraw the figures; the models and their outputs were not changed.

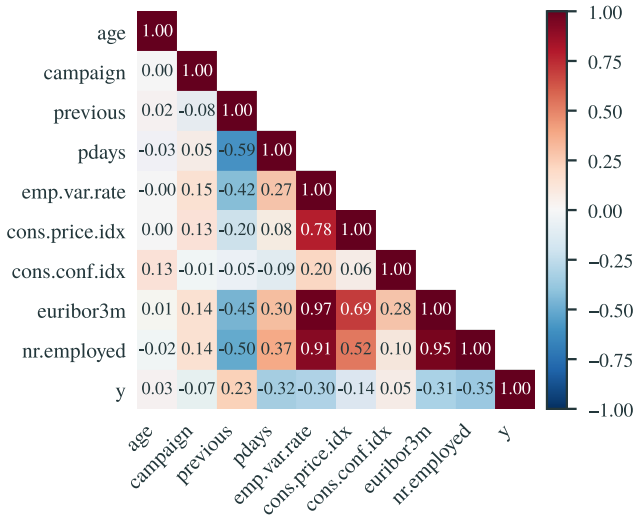


Figure 5: Pearson correlations of the numeric inputs and the outcome y.